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Additive friction stir deposition-enabled upcycling of automotive cast aluminum chips

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ABSTRACT

Additive friction stir deposition (AFSD), a deformation-based, near-net-shaping additive technology, is used to consolidate Al chips from automotive castings to produce fully-dense components, while addressing the energy, environmental, and efficiency challenges associated with recycling these chips via melting. Cold pressing of the chips results in feed-rods with a relative density of 68%. AFSD of these porous feed-rods leads to porosity-free material in the as-printed state. Compared to the base material of bulk cast Al, the as-printed material increases the tensile elongation from less than 1% to 17.8%, while exhibiting significant strain hardening. This upcycling effect is shown to originate from microstructure evolution during deposition, including the second-phase particles and the grain structure. The received Al chips have a hypereutectic composition and thus contain a high proportion of primary Si particles. The minor presence of Fe causes additional intermetallic particles as well. After deposition, these Si and Fe-based particles are refined, spherodized, and uniformly distributed in the Al matrix. Meanwhile, the cast Al microstructure is converted to an equiaxed grain structure with the grain size reduced from ~ 25 μm to ~ 2 μm. Finally, the energy consumption of AFSD-enabled upcycling is compared to melt-based recycling; potential pathways are evaluated for energy consumption reduction.

1. Introduction

Initiated with scrap processing and thermal pre-treatment, mainstream Al recycling relies on melting and solidification, wherein scraps are melted and cast into ingots. This is typically followed by rolling, extrusion, die casting, or die forging for shape control. While 95% of the energy is reduced in scrap melting compared to primary Al production, which involves Al recovery from bauxite by the Bayer process [1–3], the energy consumption and carbon footprint are still of great concern. The general industrial aluminum recycling processes (e.g., the Hoope's process and segregation process) necessitate a high working temperature of approximately 700 °C-900 °C [4,5]. Melting-based recycling is particularly inefficient for machining chips, which are characterized by a high surface area-to-volume ratio and a high oxide fraction. When Al is melted, the oxide rises to the surface and forms a layer called dross. The higher the surface area-to-volume ratio, the more dross and the lower the yield of the recycled Al. Oxidation can cause permanent material losses of 20% during remelting, leading to a low overall energy efficiency [3,6].

Given the energy, environmental, and efficiency challenges in recycling Al chips via melting, solid-state direct conversion approaches have gained attention due to their ability to reduce the recycling energy and minimize material loss while increasing the product value. Notable approaches include hot extrusion [7,8], equal channel angular pressing [9–11], friction stir extrusion [12], and friction stir consolidation [13]. The friction stir approaches are particularly effective at recycling Al chips; it is believed that the shear force results in stripping of surface oxides during processing. As a result of severe plastic deformation at elevated temperatures, the Al chips are brought into intimate contact, promoting metallurgical bond formation. In comparison to friction stir extrusion and friction stir consolidation, which produce fixed component shapes such as wires and disks, an emerging friction stir-based near-net-shaping additive process, additive friction stir deposition (AFSD) [14–18], enables free-forming and near-net-shaping, and thus holds great promise for converting waste into usable products while minimizing the number of processing steps. It has recently been demonstrated that AFSD is effective at recycling 5xxx Al [19] and Ti-6Al-4 V [20] without degrading the mechanical properties of the deposited material.

The potential of AFSD for enhancing the circular economy and materials sustainability is far from fully unlocked. Its friction stir nature not only enables material consolidation but can also modify the phase and microstructure. This possibly leads to improved mechanical prop-

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erties in the printed material compared to the base material that the chips are cut from [21], making AFSD a strong candidate for scrap *upcycling* during fabrication of 3D near-net-shaped components. To compare, there is not much upcycling effect in melt-based Al recycling because of the resulting cast microstructures and inadequate mechanical properties. While one may propose to improve mechanical properties by reformulating the alloy composition in the melt, separation of alloying elements (e.g., Mg, Zn, and Cu) from the Al matrix is extremely challenging due to its high reactivity.

In this work, the efficacy of utilizing AFSD for *upcycling* is demonstrated based on automotive machining chips made of cast Al (hypereutectic Al-Si). With simple procedures of feed-rod preparation, AFSD is shown to directly convert cast Al chips into fully dense components in the as-printed state, with the elongation to failure increasing from less than 1% to 17.8%. Thanks to significant strain hardening, the ultimate tensile strength is comparable to the base material. We show that the upcycling effect originates from the friction stir nature of AFSD, which leads to fine, equiaxed grains via dynamic recrystallization while spherodizing and reducing the size of Si and Fe-based second-phase particles. Remarkably, no heat treatment is employed in feed-rod preparation or after deposition. This discovery opens new possibilities for using cast Al chips in the automotive industry; areas of potential applications include lightweight vehicle body structures, frames, gearboxes, battery housings, etc.

2. Experimental procedure

Scrap automotive Al chips consisting of mixed Teksid Al alloys were received from Nissan Decherd Powertrain Plant in Tennessee, USA. The chips were nominally 0.1–0.3 mm thick, consisting of two in-plane geometries: ribbons (1–5 mm wide and 5–15 mm long) and plates (5 mm × 5 mm). The compositions and properties of these alloys were comparable to 3xx Al cast alloys. The as-received chips were cleaned for 100 min in an ultrasonic bath with high purity acetone, rinsed with fresh acetone, and dried. The Al chips were then compacted in a die to 68% relative density using a uniaxial press at room temperature. Pressing in this manner yielded a feed-rod with a square cross-section of 9.5 mm × 9.5 mm. Without heat treatment, the as-pressed porous feed-rods were used as the feed material for AFSD, which was carried out using a MELD R2 system (MELD Manufacturing Corporation, Christiansburg VA, USA). For demonstration purposes, a simple wall structure of 50.8 mm wide, 203.2 mm long, and 25.4 mm tall was printed at a tool rotation rate of 400 RPM, a feed rate of 1.27 mm/s, and an in-plane travel velocity of 1.27 mm/s. Each deposition track had a width of 50.8 mm and a thickness ~ 1 mm. The printed material was characterized and tested in the as-printed state without post-deposition heat treatment. Another Al alloy, AA6061-T6, with dimensions of 76 mm × 228 mm × 6.35 mm was used as the substrate (McMaster Carr, Atlanta, GA, USA).

Following ASTM-E8 for round bars, tension testing was conducted using an Instron machine (Model #4468, Canton, MA) with a 50 KN load cell. The load was oriented in the longitudinal direction of the build, which was the same as the travel direction of the print head. The elongation of the gage section was measured using an extensometer. The microstructures of the received machining chips and as-printed materials were characterized and compared via scanning electron microscopy (SEM), energy dispersive spectroscopy (EDS), and electron backscatter diffraction (EBSD), using a FEI Helios 600 NanoLab facility. The fracture surface resulting from the tension tests was examined using a JEOL IT500 HR SEM with a field emission electron source.

3. Results and discussion

The feed-rod preparation procedure and printing process presented in Fig. 1 show the as-received chips, as-pressed feed-rod, and multiple as-printed layers. Based on severe plastic deformation, AFSD is used as a near-net-shaping process with flash formation at the edges of each deposition track. After machining the rough surfaces and edges, a tensile bar with the dog-bone shape is fabricated from the resulting wall structure. Microscopy confirms that the as-printed Al component is fully dense. EDS analysis indicates that the as-received Al chips (average value of multiple chips) and as-deposited materials are very similar in composition. Both of them possess large Si content, which is common for hypereutectic Al-Si alloys. These materials are widely used for casting due to their excellent fluidity and castability, which results from the extension

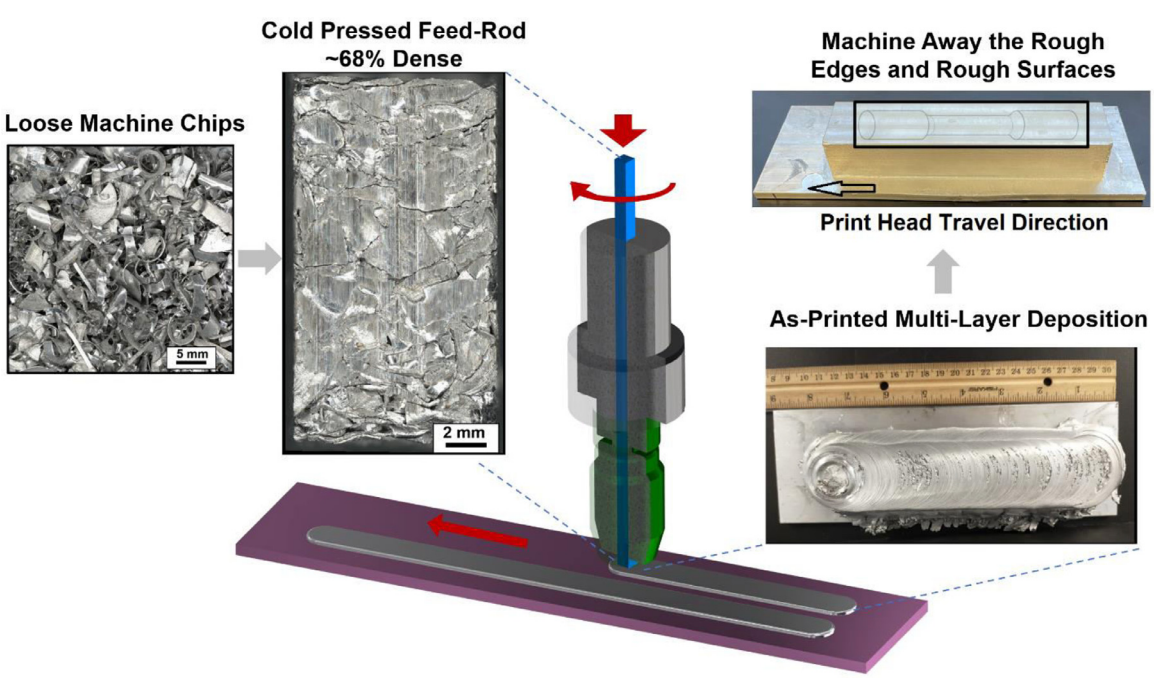

Fig. 1. Al upcycling using AFSD, showing the as-received machining chips, as-pressed porous feed-rod, as-printed multi-layers, and resulting wall structure after machining away rough edges and surfaces.

Table 1
Composition of the as-received chips and as-printed material.

Source	Element	Si	Fe	Mn	Mg	Cu	Al
Chips	wt%	15.81±1.15	0.49± 0.04	0.27±0.03	0.29±0.04	0.16±0.15	Balance
Deposition	wt%	14.06±0.23	0.42±0.06	0.26±0.05	0.27±0.02	0.3±0.05	Balance

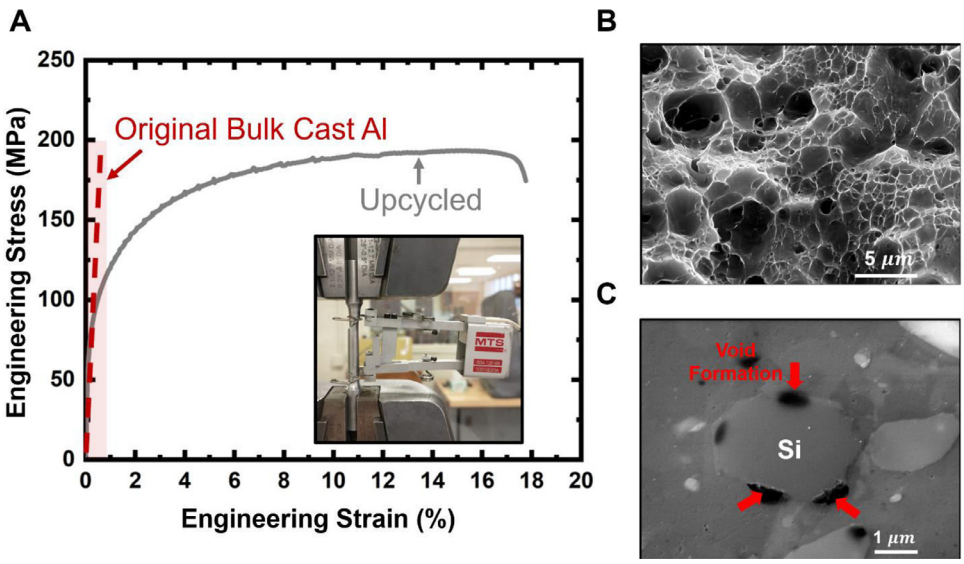

Fig. 2. (A) Tension behavior of the as-printed material using machining chips as the feedstock, with a comparison to the base material. Inset: a picture showing the fracture of a tensile bar fabricated from the as-printed material. (B) An SEM image of the fracture surface showing dimpled morphology. (C) A high magnification SEM image near the fracture surface highlighting micro-void formation around a small Si particle.

of the liquid + Si region over a wide temperature range in the phase diagram. Because of the large fraction of Si in these alloys, thermal expansion is reduced, wear resistance is improved, and the Young's modulus is increased.

Table 1 summarizes the measured compositions of the chips and deposition using EDS. This measurement is based on ten areas (nominally 1 mm × 1 mm) on the as-received Al chips and as-deposited material (five of each). At such compositions, primary Si is formed in the Al matrix upon solidification as second-phase particles, which significantly reduce the ductility and fracture toughness. In addition to the high Si composition, the presence of a minor quantity of Fe (~0.4 wt%) helps to decrease die soldering, a phenomenon where interdiffusion results in intermetallic formation and causes the casting to adhere to the mold. [22,23]. When Fe solidifies in Al-Si alloys, however, it typically forms needle-like intermetallics that act as fracture initiation sites and are detrimental to ductility [24,25]. Effectively, the hypereutectic Al-Si alloys are metal matrix composites that behave differently than conventional Al alloys.

Fig. 2(A) shows the tension testing result of as-printed material using AFSD. The material exhibits typical ductile behavior, with the 0.2% offset yield strength of 90 MPa, ultimate tensile strength of 193 MPa, and elongation to failure of 17.8%. The feed material is a blend of chips that are cut from several proprietary cast aluminum alloys developed internally by Nissan Motor Corporation and contracted suppliers. All these alloys have elongation values less than 1%; the yield strength and ultimate tensile strength are close due to the small elongation, generally in the range of 175 MPa to 200 MPa. Detailed information regarding metal composition and mechanical properties is classified as trade secret and exempt from public disclosure. The area under the stress-strain curve corresponds to the energy absorption per volume; it is a measure of material toughness. To compare, the energy absorption per volume is 3.1 $\times 10^7$ J/m^3 for the as-printed material, and is ~ 1.0 $\times 10^6$ J/m^3 for the base material. This represents an increase of over 30 times.

It is noted that the results presented here do not include the out-of-plane tensile properties. In typical metal additive manufacturing processes like powder bed fusion and directed energy deposition, mechanical anisotropy is commonly observed due to their layer-by-layer nature. Such anisotropy is expected to be substantially reduced in AFSD thanks to the vertical material flow across layers; our ongoing work on AFSD of Ti-6Al-4 V shows that the in-plane and out-plane tensile properties can be comparable. Understanding the mechanical anisotropy in AFSD-enabled recycling can be an important direction in future exploration.

As shown in the inset of Fig. 2(A), the tensile bar made from the as-printed material fails along a 45-degree plane within the gage section, suggesting shear-dominated deformation and failure, which is typical for ductile materials. Examination of the fracture surface in SEM reveals dimple morphology with two distinct dimple sizes, suggesting that the failure is caused by micro-void coalescence (see Fig. 2(B)). Fig. 2(C) shows a zoomed-in image of a cross section near the fracture surface, highlighting a small Si particle with micro-voids formed around it. This indicates that the interfaces between the Si particles and Al matrix could be important sites for initiating micro-void coalescence. This contrasts with the failure mode of conventional cast Al-Si alloys, which is characterized by the breakage of Si particles and intermetallics [26]. The particle breakage provides crack initiation sites, and the cracks can easily propagate along near-continuous paths through debonding of the particle-matrix interface. Such failure is exasperated by the elongated shape of the particles, which is common in hypereutectic Al-Si.

For general structural applications that consider toughness and elongation as the key metrics, AFSD-based recycling results in drastic improvement from the base material, which is the brittle bulk cast Al, resulting in an 'upcycling' effect. To understand the origin of the upcycling effect, the microstructures of the as-received cast Al chips and as-printed materials are characterized and compared, as detailed in Fig. 3. SEM characterization is performed on multiple chips, where the presence of both primary Si and Fe-containing intermetallic particles is discovered. The volume fraction and morphology of these particles are similar among different chips. Representative results are shown in Fig. 3(A) and (C) at different magnifications. In these images, the bright features contain Fe, the dark features are Si, and the matrix is Al. Fe-containing intermetallics are shown to vary in morphology, including (i) needle-like shapes with widths around 3 μm and lengths up to 100 μm; (ii) large globular shapes with a block-like appearance on the order of 10 μm; (iii)

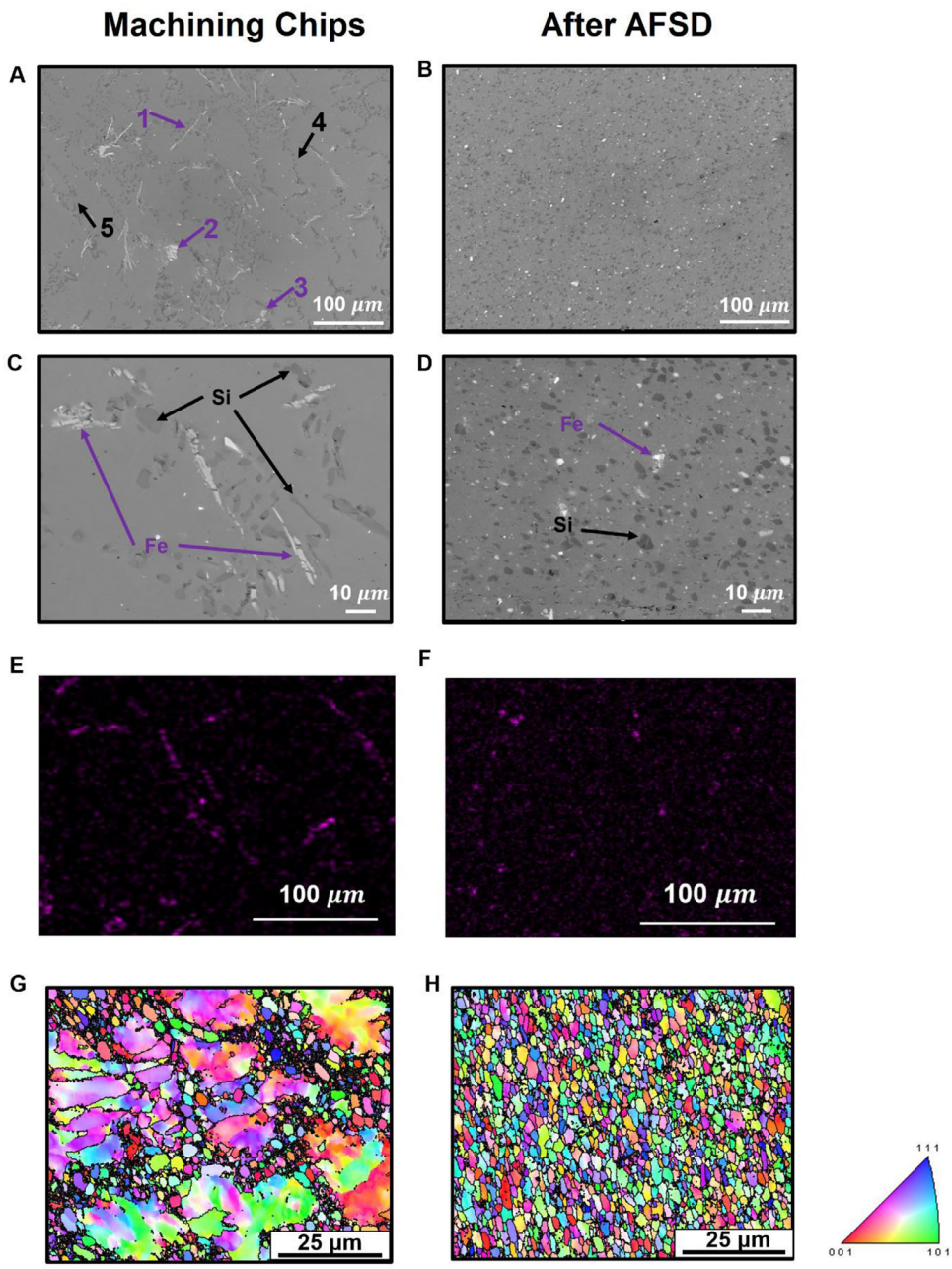

Fig. 3. Comparison of the microstructure and phase between as-received Al machining chips (left column) and as-printed materials (right column) by AFSD. This includes (A)-(D) SEM images, (E) and (F) EDS mapping of Fe, and (G) and (H) inverse pole figure maps based on EBSD.

small, equiaxed shapes nominally 5 μm in diameter. These three types are labeled '1′, '2′, and '3′in Fig. 3(A), respectively. Overall, the needle-like shape is prevalent, whereas the other two types are much less common. In the machining chips, the primary Si particles also exhibit multiple morphologies. The most common includes equiaxed shapes with diameters ranging from 1 to 10 μm, while less common shapes are plate-like with the length up to 40 μm. Interestingly, whether equiaxed or plate-like in shape, the primary Si particles tend to be along the grain boundaries, appearing as near-continuous pathways on the macroscopic level, as highlighted in Fig. 3(A) (labeled '4′ and '5′).

The as-printed material shows a stark contrast in the morphology and distribution of Fe-containing and primary Si particles. As shown in Fig. 3(B) and (D), both Fe and Si particles have been refined to a size mostly in the range of 1–3 μm. In addition, a higher fraction of submicron particles is observed compared to that in the chips. Importantly, these second-phase particles are all globular or equiaxed with no elongated geometries. Fig. 3(E) and (F) compare the Fe element mapping between the machining chips and as-printed material, highlighting the size reduction and shape modification through AFSD. The drastic changes of the second-phase particles originate from shear forces imposed by the print head during AFSD. The deposition temperature is approximately 60%–90% of the melting temperature for Al matrix [27], which reduces its viscosity and undergoes viscoplastic flow. This temperature, however, is still low for the Fe-containing and primary Si particles, which exhibit brittle behavior during deposition and fracture under shear stress. As observed in other friction stir-related processes [28,29], this leads to simultaneous spheroidization and particle size reduction. Furthermore, the stir operation rooted in AFSD aids in uniformly dispersing these small second-phase particles throughout the Al matrix, which can clearly be seen in Fig. 3(B) and (D).

Fig. 3(G) and (H) compare the grain structures of the Al machining chips and as-printed materials based on inverse pole figure maps. In the machining chips, the Al grains exhibit irregular shapes and are on the order of ~25 μm. The grains are clearly deformed, as evidenced

Table 2
Comparison between AFSD and melt-based recycling for cast Al machining chips.

	Melt-based Al recycling	AFSD-based Al recycling
Scrap processing	Scrap can be melted directly after cleaning	Scrap needs to be formed into feed-rods after cleaning
Oxides & impurities	Dross removal and virgin material addition are necessary to fix composition	High tolerance to oxides and impurities
Bonding	Liquid phase bonding by melting and solidification	Solid-state bonding by deformation and diffusion
Shape control	Secondary processing, either remelting, or extrusion/rolling is required after ingot formation to make products	Near-net-shaped products can be made directly
Grain structure	Cast microstructure	Wrought microstructure
Second-phase particle: shape	Needle-shaped iron phases	Spherical iron phases (lower stress concentrators)
Second-phase particle: spatial distribution	Near-continuous silicon pathways	Evenly distributed silicon
Mechanical properties	Poor ductility	Excellent ductility

by the spectrum of orientations present in individual grains. This high level of deformation results from the machining process. In contrast, the as-printed materials consist of fine, equiaxed Al grains with an average size of 2 μm. The grain size reduction stems from dynamic recrystallization during AFSD. The uniform grain size distribution suggests that continuous dynamic recrystallization likely plays an important role, which has also been observed in AFSD of 7xxx, 6xxx, and 2xxx Al alloys [14-16, 30-35]. Poorly indexed regions in EBSD analysis correspond to the second-phase particles and appear in black. In the machining chips, primary Si particles are distinguishable from the Al grains, as the latter are much larger. Consistent with the SEM observation, Si particles are concentrated around Al grain boundaries, making near-continuous paths. In the as-printed materials, the second-phase particle size is comparable to or smaller than the Al grain size.

From Fig. 3, it is concluded that the upcycling effect in AFSD of cast Al chips primarily originates from the microstructure changes. In the base material, the presence of large, needle-like Fe-containing intermetallic particles is reported to reduce the ductility [24], whereas the Si particles along the grain boundaries promote intergranular fracture. After AFSD, the fine, globular Fe and Si-based particles are uniformly distributed in the Al matrix, preventing the material from brittle failure under tension. Due to the difference in elastic modulus between the matrix material and particles, the interface between the two represents a strain discontinuity; geometrically necessary dislocations accumulate in the interfacial regions under stress, resulting in strain hardening. This can give rise to a substantial increase in the ultimate tensile strength from the yield strength, as shown in Fig. 2(A). With sufficiently large deformation, micro-voids are initiated at these interfaces, growing and connecting with other voids. This finally results in the dimple morphology of the fracture surface. Evolution of the grain structure also helps to improve the ductility. In the base material, primary Si particles accumulate along the boundaries of coarse grains, providing near continuous paths that cause brittle fracture. After AFSD, the size of recrystallized grains becomes comparable to the second-phase particles, which are uniformly dispersed in the matrix.

The yield strength decrease after AFSD is caused by the decrease in dislocation density. In cast Al alloys, the different coefficients of thermal expansion between Al and Si cause high dislocation content around the interface, leading to quench strengthening [36]. After AFSD, the dislocation density significantly decreases in the Al matrix, e.g., through dislocation recovery and reorganization in the presence of dynamic recrystallization and particle breakage. Although grain size and particle size are reduced after AFSD, the weak Hall-Petch strengthening in Al [37] and the weak Orowan strengthening associated with micro-particles are still unable to compensate for the decline of yield strength caused by dislocation reduction.

Table 2 compares AFSD-based and melt-based Al recycling for cast Al machining chips, as a function of processing, microstructure, and properties.

Finally, a basic energy analysis is conducted for AFSD-based upcycling and compared with melt-based Al recycling, with a focus on the on-site energy consumption. For AFSD-based upcycling of Al chips, the total energy input per mass E_M can be written as $E_M = P/(F \times A_{Feed-rod} \times \rho)$, where P is the operation power, F is the feed rate denoting the vertical velocity for feeding the material, $A_{Feed-rod}$ is the cross-sectional area of the feed-rod, and ρ is the feed material density. $F \times A_{Feed-rod} \times \rho$ represents the deposition rate, with a unit of mass per time. In this work, $P = 3.5\ kW$, $F = 1.27\ mm/s$, $A_{Feed-rod} = 9.525 \times 9.525 = 90.73\ mm^2$, $\rho = 68\% \times 2700\ kg/m^3$ (recall that the relative density of the as-compressed material is 68%). These numbers yield $E_M = 4.63\ kWh/kg$ and a deposition rate of 0.76 kg/hr (see Fig. 4).

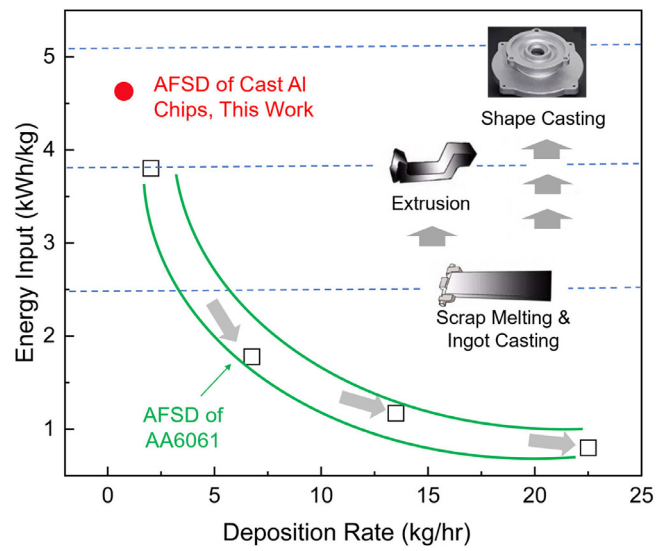

Fig. 4. A comparison of energy input per material mass among multiple processes, including AFSD of cast Al chips shown in this work, AFSD of AA6061 using solid bars, and melt-based Al recycling involving ingot casting, extrusion, and shape casting.

For melt-based Al recycling, it costs approximately 2.50 kWh/kg in scrap melting and ingot casting [1]. Ingot casting is not as intrinsically energy intensive, typically represented by a batch process with large quantities of molten metal held in furnaces for alloying, fluxing, and degassing. Ingots are then subjected to rolling, extrusion, die casting, or die forging for shape control. The additional energy cost is 1.30 kWh/kg for extrusion and 2.56 kWh/kg for die-casting or shape-casting. Considering the total energy cost, a value of 3.80–5.06 kWh/kg is estimated to form shapes by extrusion or die casting following scrap melting and ingot casting. This range is comparable to the aforementioned energy cost calculated for AFSD-based upcycling of Al chips ($E_M = 4.63\ kWh/kg$).

With the above analysis, AFSD-based Al upcycling indicates no apparent advantage in energy consumption when compared to traditional melt-based Al recycling (see Fig. 4). This conclusion appears to be counterintuitive at first glance, as the former is a solid-state process without material melting. The reason behind this conclusion is that melt-based Al recycling is a mature, large-scale process that has evolved over time to use less energy per unit mass. In contrast, the use of AFSD for material recycling is still in the early research stage, and the demonstrated exper-

iments are performed at a much smaller scale with a very low deposition rate of 0.76 kg/hr. The authors expect that a higher deposition rate will lead to reduction of energy input per mass in AFSD-enabled upcycling; this hypothesis needs to be tested in future experiments. Encouragingly, a similar trend has been observed in AFSD of AA6061 using solid bars as the feed material (see Fig. 4). As the deposition rate of AA6061 increases 11 times from 2 kg/hr to 22 kg/hr, the operation power only increases 2.3 times, from 7.7 kW to 18 kW, so the energy consumption per mass drops from 3.8 kWh/kg to 0.8 kWh/kg.

4. Conclusions

To summarize, it is demonstrated that AFSD enables simultaneous near-net-shape deposition and upcycling of cast Al chips, in which cold-pressed porous feed-rods are employed to result in fully dense deposition. The tensile elongation increases from less than 1% in the bulk base material to 17.8% in the as-printed material with significant strain hardening. AFSD features severe shear deformation along with salient phase and microstructure evolution. The upcycling effect is mainly caused by the size reduction, spheroidization, and dispersion of Si and Fe-based second-phase particles that are initially present in the base material and machining chips. The energy consumption for AFSD-based upcycling in this work is comparable to melt-based Al recycling, which involves scrap melting and ingot casting, followed by extrusion or shape casting. Future work aiming to increase the deposition rate in AFSD will likely lower the energy consumption per mass.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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